How Do We Formulate Erasure Using Statistical Independence ?

$$Z_\theta = \varepsilon(X; \boldsymbol{\theta}) \quad \rightarrow$$

$Z_\theta \rightarrow$ Random Variable After Erasure

$S \rightarrow$ Random Variable Represent Unwanted Label

$Y \rightarrow$ Random Variable Represent Utility Label

$$\inf_\theta \quad \mathbf{HSIC}(Z_\theta, S) - \mathbf{HSIC}(Z_\theta, Y)$$

Conflicting Objectives $\equiv$ Challenging Optimization

Minimize Statistical Dependency

A Proxy to Preserve Utility Information

How Do We Solve This Optimization ?

Single-Shot Optimization Leads to Poor Solutions